



Habib University

CS-330 Computer Architecture – Fall 2025

Instructor: Dr. Shehzad Hasan

Time = 50 minutes

Quiz 01

Max Points: 20

Instructions:

- i. **Smart watches, laptops, and similar electronics are strictly NOT allowed.**
- ii. **Answer sheets should contain all steps, working, explanations, and assumptions.**
- iii. Attempt the quiz on clean papers with black/blue ink.
- iv. Print your name and HU ID on all sheets.
- v. Turn in your question paper along with your answer sheets.
- vi. You are not allowed to ask/share your method or answer with your peers. The work submitted by you is solely your own work. Any violation of this will be the violation of HU Honor code and proper action will be taken as per university policy if found to be involved in such an activity.

CLO Assessment:

This assignment assesses students for the following course learning outcomes.

Course Learning Outcomes		CLO Assessed
CLO 1	<i>Explain</i> the role of ISA in modern processors and instruction encodings and assembly language programming	✓
CLO 2	<i>Explain</i> the architecture and working of a single cycle processor	
CLO 3	<i>Design</i> the architecture to mitigate issues of a pipelined processor	
CLO 4	<i>Analyze the</i> performance of cache operations	

Question 1 [4 points]:

Compilers can have a profound impact on the performance of an application. Suppose there are two compilers A and B

- a. Assume that for an application, compiler-A generates a code having 5×10^9 dynamic instructions which are executed in 9.78 s, while compiler-B generates a code having 2.35×10^9 dynamic instructions with an execution time of 4.15 s on the same processor. Find the average *CPI* for each program given that the processor has a clock rate of 1.1 GHz. (2 marks)

Since $CPU\ time = Instruction\ count \times CPI_{avg} \times Clock\ cycle\ time$
Therefore, $CPI_{avg} = CPU\ time / (Instruction\ count \times Clock\ cycle\ time)$
or, $CPI_{avg} = CPU\ time \times Clock\ rate / Instruction\ count$

For Compiler A: $CPI_{avg} = 9.78 \times 1.1 \times 10^9 / (5 \times 10^9) = 2.1516$

For Compiler B: $CPI_{avg} = 4.15 \times 1.1 \times 10^9 / (2.35 \times 10^9) = 1.9426$

- b. Assume the compiled programs now run on two different processors (compiler-A's code on P1 and compiler-B's code on P2). If the execution times on the two processors are the same, which processor's clock is faster and by how much? (2 marks)

Now we are going to run these programs on different machines having different clock times. The CPI_{avg} as computed earlier is not going to change as it represents the set of different classes of instructions and their cycles which is not going to change if we run them on a different processor.

In order for the CPU times to be same

$$\frac{Instruction\ count_A \times CPI_{avg}A}{clock\ rate_P1} = \frac{Instruction\ count_B \times CPI_{avg}B}{clock\ rate_P2}$$

$$\frac{5 \times 10^9 \times 2.1516}{clock\ rate_A} = \frac{2.35 \times 10^9 \times 1.9426}{clock\ rate_B}$$

$$\frac{clock\ rate_A}{clock\ rate_B} = \frac{5 \times 10^9 \times 2.1516}{2.35 \times 10^9 \times 1.9426}$$

$$\frac{clock\ rate_A}{clock\ rate_B} = 2.357$$

Clock of the processor P1 running compiler A's code is 2.357 times faster.

Question 2 [4 points]:

Consider the following program

```
addi    x4, x0, 1000
addi    x5, x0, 100
loop:   bge    x5, x4, exit
        slli   x6, x5, 2
        lwu    x7, 0(x6)
        sw     x7, 100(x6)
        addi   x5, x5, 4
        jal    x0, loop
exit:
```

Let the clock cycle counts for each instruction type be as follows:

Arithmetic: 2 cycles

Logical: 1 cycle

Memory: 4 cycles

Conditional branching: 3 cycles

Unconditional branching: 2 cycles

Calculate the total number of clock cycles required to execute the program based on the instruction distribution. Let the instructions in the loop run 25 times, the loop terminating instruction, however, is going to run one more time.

Total instructions: $2A + 25(1A + 1L + 2M + 1C + 1U) + 1C = 153$ instructions

Total Clock Cycles: $2 \times 2 + 25(1 \times 2 + 1 \times 1 + 2 \times 4 + 1 \times 3 + 1 \times 2) + 1 \times 3 = 407$ cycles

Question 3 [4 points]:

Consider a computer running a program that requires 350s, with 140s spent executing memory instructions, and 120s spent executing arithmetic instructions.

- a. What would be the speedup achieved if the memory unit is enhanced by a factor of 5. **(2 marks)**

Of the total program time only 140s could be enhanced by a factor of 5.

Time that cannot be enhanced = $350 - 140 = 210$ s

Time after enhancement = $210 + 140/5 = 180 + 14 = 238$ s

Speedup = $350/238 = 1.47$

- b. Can the total program time be reduced by 20% by reducing only the time for arithmetic instructions? If yes, then by what factor should the arithmetic unit be improved? **(2 marks)**

Arithmetic instructions take 120s of the total program time.

Time that is not enhanced = $350 - 120 = 230$ s

Time after 20% reduction of 350 s = $0.8 \times 350 = 280$ s

Since this time is greater than the time that is not enhanced then yes the program time can be reduced by 20% by just improving the arithmetic unit.

Factor by which it should be enhanced

$230 + 120/x = 280$

Therefore $x = 120/50 = 2.4$

So arithmetic unit has to be improved 2.4 times.

Question 4 [4 points]:

- a. Assume that array A is an array of 100 double words, x20 contains the base address of array A, and x21 corresponds to variable D. Compile the following C expression into its equivalent RISC-V assembly instructions **(2 marks)**

$$A[100] = D + A[18] - 30$$

```
ld    x5, 144(x20)
addi  x5, x5, -30
add   x5, x5, x21
sd    x5, 800(x20)
```

- b. Consider now array A is an unsigned array of 100 halfwords. Make the required changes in the RISC-V assembly instructions for the same C expression. **(2 marks)**

```
lhu   x5, 36(x20)
addi  x5, x5, -30
add   x5, x5, x21
sh    x5, 200(x20)
```

Question 5 [4 points]:

Encode the following RISC-V instructions to machine language code

- a. lhu x9, -1396(x27)

-1396 = 100010011000

																											opcode									
1	0	1	0	1	0	0		1	1	0	0	0		1	1	0	1	1		1	0	1		0	1	0	0	1		0	0	0	0	0	1	1

- b. sll x19, x21, x20

																						opcode								
0	0	0	0	0	0	0	0	1	0	1	0	0	1	0	1	0	0	1	1	0	0	1	1	0	1	1	0	0	1	1